

BRANDON LUM

☎ (415) 568-0884 | ✉ brandon.lum650@gmail.com | 🐙 github.com/b-lum | 💻 [Website](#) | 🔗 linkedin.com/in/b-lum

EDUCATION

University of California San Diego

Bachelor of Science in Data Science / GPA: 3.41

Sep 2025 – Jun 2028

La Jolla, CA

College of San Mateo

A.S. in Computer and Information Science, A.S. in Mathematics / GPA: 3.86

Aug 2024 – May 2025

San Mateo, CA

Relevant Coursework: Data Structures & Algorithms, Intro to Data Management, Data Visualization, ML: Learnign Algorithms

EXPERIENCE

Software Engineer Intern

SkyAccess

Jun 2026 – Sep 2026

Beverly Hills, CA

- Analyzed 6.3K+ bookings to prove 62% vs 19% quote error, shipped a production RCT, and shelved a feature on zero demand
- Proposed and owned a company-wide pricing-engine surge detector end-to-end, shipping 5 phases dark behind feature flags
- Implemented passwordless WebAuthn with an atomic Redis Lua challenge store and IDOR-safe endpoints, unit and E2E tested
- Audited 8.8K canceled bookings into a mutually exclusive taxonomy, catching 3 spec defects and a dead outcome field
- Shipped 6 features across 16 PRs in 2 production TypeScript repos (Express, Prisma, Postgres, Redis, Next.js, Turborepo)

Software Development Engineer Intern

DeepSpace

Jan 2026 – Jun 2026

San Diego, CA (Remote)

- Shipped and owned 4+ web applications end-to-end (TypeScript, React, Next.js, Express) at a pre-launch AI-native startup
- Integrated and maintained 6+ 3rd-party services across production apps, ensuring reliable pipelines and API functionality
- Conducted peer reviews for 3+ engineers' project applications, providing feedback on code quality and architecture

Undergraduate Researcher

Stanford University

Sep 2024 – Apr 2025

Palo Alto, CA

- Researched ML image classification for chronic renal disease, reviewing 4 studies to evaluate model effectiveness
- Built a NumPy/SciPy pipeline over HWiNFO64 logs sampled every 2s, cleaning CPU and GPU telemetry across 6 test runs
- Ran Welch's t-tests on an added intake fan: CPU temps fell 1.3°C ($p < 0.001$) while GPU deltas stayed insignificant
- Refined methodology weekly with 5 researchers and presented the null GPU result at a Stanford SSG spring research talk

Center Assistant

Kumon Math and Reading Center

Sep 2021 – Aug 2025

San Bruno, CA

- Instructed K-12 students in math through college-level calculus, leading the Math section and covering ELA as needed
- Ran front-desk operations and fielded customer inquiries for 50 families daily, and trained 5+ new hires on center workflows

PROJECTS

Dragon Boat Team Portal | *Next.js, TypeScript, Supabase, Postgres, RLS, Tailwind, MapLibre GL*



- Built and run the ops portal a 50-person team uses weekly: attendance forms, boat lineups, and carpool assignment
- Enforced multi-team isolation in Postgres with Supabase row-level security across 8 migrations, RLS policies, and RPCs
- Cut signup friction with passwordless email sign-in and anonymous shared form links, restoring form response rates
- Shipped weight-balanced lineup autofill and carpool assignment as unit-tested packages in a pnpm monorepo

Carpool Optimizer | *React, Vite, Express, MapLibre GL, OSRM, OpenStreetMap*

🖥️ Deployment 🐙

- Cut student club ride-planning time an estimated 30% with automated seat assignment and pickup-order route optimization
- Implemented greedy nearest-neighbor assignment preserving manual seat picks while respecting per-car capacity and absences
- Rendered per-car color-coded routes in MapLibre GL over self-hosted OSRM, with a runtime toggle to the public OSRM API
- Shipped a companion Python CLI with pickup and drop-off modes plus an offline no-routing path, installable directly via uv

F1 Telemetry + Oversteer Detection | *Python, PyTorch, Hugging Face, PEFT, FastAPI, React*

🐙 App 🐙 Model

- Built a FastAPI WebSocket service that unpacks live F1 2020 UDP telemetry into speed, throttle, brake, and per-wheel slip
- Decoupled ingest from render with a client packet queue drained at 20 FPS, keeping Chart.js smooth without React state thrash
- Fine-tuned VideoMAE with LoRA to classify oversteer, understeer, and neutral over 972 labeled 60-frame driving clips
- Interpolated positional embeddings from 16 to 60 frames and cut trainable params to 1.3M of 88M via LoRA on query/value

AREAS OF EXPERTISE

Skills: Java, Python, TypeScript, SQL, JavaScript, HTML/CSS, React, Next.js, Node.js, Express, PostgreSQL, Redis, Prisma, Bull, node-cron, Turborepo, JavaFX, Kivy, PyTorch, Scikit-learn, Statsmodels, Pandas, NumPy, SciPy, Matplotlib, Seaborn, PEFT, RAG, LlamaIndex, REST APIs, WebSocket APIs, Git, Unix, Docker, Hugging Face, Supabase, Vercel, Cloudflare, MongoDB